Ch29：6，7,16，19，21，24，26，27

Basic score 4` + problem 6 2`+ problem 21 2`+ problem 26 2`

6. (a) The contribution to B_C from the (infinite) straight segment of the wire is

$$B_{C1} = \frac{\mu_0 i}{2\pi R}. \quad \text{0.5 point}$$

The contribution from the circular loop is $B_{C2} = \frac{\mu_0 i\phi}{4\pi R} = \frac{\mu_0 i 2\pi}{4\pi R} = \frac{\mu_0 i}{2R}$ 0.5 point

Thus,

$$B_C = B_{C1} + B_{C2} = \frac{\mu_0 i}{2R}\left(1+\frac{1}{\pi}\right) = \frac{(4\pi\times10^{-7}\,\text{T}\cdot\text{m/A})(5.78\times10^{-3}\,\text{A})}{2(0.0154\,\text{m})}\left(1+\frac{1}{\pi}\right) = 3.11\times10^{-7}\text{T}.$$

$\vec{B}_C$ points out of the page, or in the +z direction. In unit-vector notation,

$$\vec{B}_C = (3.11\times10^{-7}\text{T})\hat{\text{k}}$$ 0.5 point 公式 0.3 point，结果 0.2 point

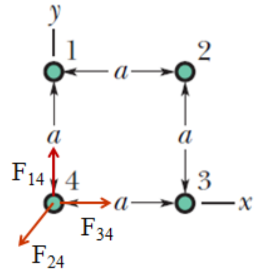
(b) Now, $\vec{B}_{C1} \perp \vec{B}_{C2}$ and B_{C1} is still in z direction but B_{C2} is in x direction now, so

$$B_C = \sqrt{(2.36\times10^{-7}\text{T})^2 + (0.75\times10^{-7}\text{T})^2} = 2.48\times10^{-7}\ \text{T}$$

$$\vec{B}_C = \left(\frac{\mu_0 i}{2R}\right)\hat{\text{i}} + \left(\frac{\mu_0 i}{2\pi R}\right)\hat{\text{k}} = (2.36\times10^{-7}\text{T})\hat{\text{i}} + (7.52\times10^{-8}\text{T})\hat{\text{k}}.$$

0.5 point 公式 0.3 point，结果 0.2 point

7. We use the superposition of forces: $\vec{F}_4 = \vec{F}_{14} + \vec{F}_{24} + \vec{F}_{34}$. With $\theta = 45°$, the situation is as shown on the right. The components of $\vec{F}_4$ are given by



$$F_{4x} = F_{34} - F_{24}\cos\theta = \frac{\mu_0 i^2}{2\pi a} - \frac{\mu_0 i^2 \cos 45°}{2\sqrt{2}\pi a} = \frac{\mu_0 i^2}{4\pi a}$$

and

$$F_{4y} = F_{41} - F_{42}\sin\theta = \frac{\mu_0 i^2}{2\pi a} - \frac{\mu_0 i^2 \sin 45°}{2\sqrt{2}\pi a} = \frac{\mu_0 i^2}{4\pi a}.$$

Thus, in unit-vector notation, we have

$$\vec{F}_1 = \frac{\mu_0 i^2}{4\pi a}\hat{\text{i}} + \frac{\mu_0 i^2}{4\pi a}\hat{\text{j}} = 4.17\times10^{-5}\hat{\text{i}} + 4.17\times10^{-5}\hat{\text{j}}\ \ N$$

16. As the problem states near the end, some idealizations are being made here to keep the calculation straightforward (but are slightly unrealistic). For circular motion (with speed, $v_\perp$, which represents the magnitude of the component of the velocity perpendicular to the magnetic field, the period is (see Eq. 28-17)

$$T = 2\pi r/v_\perp = 2\pi m/eB.$$

Now, the time to travel the length of the solenoid is $t = L/v_\parallel$ where $v_\parallel$ is the component of the velocity in the direction of the field (along the coil axis) and is equal to $v\cos\theta$, where θ = 30°. Using Eq. 29-23 ($B = \mu_0 in$) with $n = N/L$, we find the number of revolutions made is:

$$\frac{t}{T} = \frac{L}{v_\parallel}\frac{eB}{2\pi m} = \frac{L}{v_\parallel}\frac{e\mu_0 Ni}{2\pi mL} = \frac{e\mu_0 Ni}{2\pi mv\cos 30°} = 2.6\times10^6$$

19. Consider a section of the ribbon of thickness dx located a distance x away from point P. The current it carries is $di = i\,dx/w$, and its contribution to B_P is

$$dB_P = \frac{\mu_0 di}{2\pi x} = \frac{\mu_0 i dx}{2\pi xw}.$$

Thus,

$$B_P = \int dB_P = \frac{\mu_0 i}{2\pi w}\int_d^{d+w}\frac{dx}{x} = \frac{\mu_0 i}{2\pi w}\ln\left(1+\frac{w}{d}\right) = \frac{(4\pi\times10^{-7}\,\mathrm{T\cdot m/A})(4.61\times10^{-6}\,\mathrm{A})}{2\pi(0.062\,\mathrm{m})}\ln\left(1+\frac{0.062}{0.0161}\right)$$
$$= 2.35\times10^{-11}\mathrm{T}.$$

and $\vec{B}_P$ points upward. In unit-vector notation, $\vec{B}_P = (2.23\times10^{-11}\mathrm{T})\hat{\mathrm{j}}$

Note: In the limit where $d \gg w$, using

$$\ln(1+x) = x - x^2/2 + \cdots,$$

the magnetic field becomes

$$B_P = \frac{\mu_0 i}{2\pi w}\ln\left(1+\frac{w}{d}\right) \approx \frac{\mu_0 i}{2\pi w}\cdot\frac{w}{d} = \frac{\mu_0 i}{2\pi d}$$

which is the same as that due to a thin wire.

21. The magnitudes of the forces on the sides of the rectangle that are parallel to the long straight wire (with i_1 = 30.0 A) are computed using Eq. 29-13, but the force on each of the sides lying perpendicular to it (along our y axis, with the origin at the top wire and $+y$ downward) would be figured by integrating. Fortunately, these forces on the two perpendicular sides of length b cancel out. For the remaining two (parallel) sides of length L, we obtain

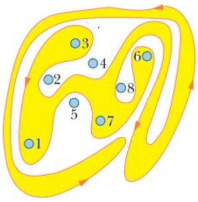
$$F = \frac{\mu_0 i_1 i_2 L}{2\pi}\left(\frac{1}{a} - \frac{1}{a+b}\right) = \frac{\mu_0 i_1 i_2 bL}{2\pi a(a+b)}$$ 1 point（0.5 point for each section of wire）

$$=\frac{\left(4\pi\times10^{-7}\,\mathrm{T\cdot m/A}\right)(30.0\mathrm{A})(20.0\mathrm{A})(8.00\mathrm{cm})(20\mathrm{cm})}{2\pi(1.00\mathrm{cm})(1.00\mathrm{cm}+8.00\mathrm{cm})}=2.13\times10^{-3}\,\mathrm{N},$$ 0.5 point

and $\vec{F}$ points toward the wire, or $+\hat{\mathrm{j}}$. That is：

$\vec{F}=(2.13\times10^{-3}\,\mathrm{N})\hat{\mathrm{j}}$ in unit-vector notation. 0.5 point

24. A close look at the path reveals that only currents 1, 3, 6 and 7 are enclosed. Thus, noting the different current directions described in the problem, we obtain



$$\oint\vec{B}\cdot d\vec{s}=\mu_0\left(7i-6i+3i+i\right)=5\mu_0 i$$
$$=5\left(4\pi\times10^{-7}\,\mathrm{T\cdot m/A}\right)\left(6.00\times10^{-3}\,\mathrm{A}\right)=3.77\times10^{-8}\,\mathrm{T\cdot m}.$$

26. (a) The field at the center of the pipe (point C) is due to the wire alone, with a magnitude of：

$$B_C=\frac{\mu_0 i_{\text{wire}}}{2\pi(3R)}=\frac{\mu_0 i_{\text{wire}}}{6\pi R}.$$ 0.5 point

For the wire we have $B_{P,\text{ wire}} > B_{C,\text{ wire}}$. Thus, for $B_P = B_C = B_{C,\text{ wire}}$, i_{wire} must be into the page:

$$B_P=B_{P,\text{wire}}-B_{P,\text{pipe}}=\frac{\mu_0 i_{\text{wire}}}{2\pi R}-\frac{\mu_0 i}{2\pi(2R)}.$$ 0.5 point

Setting $B_C = -B_P$, 0.3 point

We obtain $i_{\text{wire}} = 3i/8 = 3(2.12\times10^{-3}\,\mathrm{A})/8=7.95\times10^{-4}\,\mathrm{A}$. 0.2 point

(b) The direction is into the page. 0.5 point

27. (a) We denote the $\vec{B}$ fields at point P on the axis due to the solenoid and the wire as $\vec{B}_s$ and $\vec{B}_w$, respectively. Since $\vec{B}_s$ is along the axis of the solenoid and $\vec{B}_w$ is perpendicular to it, $\vec{B}_s\perp\vec{B}_w$. For the net field $\vec{B}$ to be at 45° with the axis we then must have $B_s = B_w$. Thus,

$$B_s=\mu_0 i_s n=B_w=\frac{\mu_0 i_w}{2\pi d},$$

which gives the separation d to point P on the axis:

$$d=\frac{i_w}{2\pi i_s n}=\frac{5.00\,\mathrm{A}}{2\pi\left(35.0\times10^{-3}\,\mathrm{A}\right)(10\,\mathrm{turns/cm})}=2.27\,\mathrm{cm}.$$

(b) The magnetic field strength is

$$B = \sqrt{2}B_s = \sqrt{2}\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(35.0\times10^{-3}\ \text{A}\right)\left(10\ \text{turns}/0.0100\ \text{m}\right) = 6.22\times10^{-5}\,\text{T}.$$